The Project

This report analyzes customer churn for a subscription-based business using a dataset of ~440,000 customer records sourced from Kaggle. While the data is publicly available for portfolio and learning purposes, the analytical framework such as segmentation logic, risk flagging, cost-to-serve modeling, is built to reflect the decisions a real retention or customer strategy team would face.

The Dataset

Source	Customer Churn Dataset
File type	CSV
Rows	~440,000
Original columns	12
After transformation	24+
Behavioral window	30-day snapshot
Spend field	Cumulative lifetime spend (not periodic revenue)
Model structure	Single-table
Source	customer-churn-dataset

Business Scenario

A subscription business is experiencing high churn and needs to understand which customers are leaving, why, and what it costs the organization. Leadership wants to move from reactive reporting to proactive retention by identifying at-risk customers before they cancel, protecting high-value segments, and making smarter decisions about where to invest support and marketing resources.

Questions Answered by Business Roles

Executive / C-Suite

- Where is revenue concentrated, and how exposed are we if those customers churn?
- Which customer segments are generating outsized operational cost relative to their value?
- What does a healthy vs. at-risk customer look like?

Customer Success / Retention

- Which customers are most likely to churn in the near term?
- What behavioral signals appear before a customer cancels?
- Which segments should be prioritized for proactive outreach?

Marketing

- What does the highest-value customer profile look like and and how do we attract more of them?
- Are there plan types, tenure bands, or spend tiers that correlate with long-term retention?
- Which segments are underperforming and may need different messaging or incentives?

Support / Operations

- Which customer segments are generating disproportionate support volume?
- Is high support usage a churn signal, a service quality issue, or both?
- Where can support resources be better allocated or gated?

Product

- Which plan types retain customers longest, and which see the highest churn?
- Are there tenure or activity patterns that suggest product-fit problems?
- What does usage look like across the customer lifecycle?

Report Structure

Section	Focus
1-3	Exploratory analysis: demographics, behavior, account attributes
4	Cost to serve: revenue vs. support burden by segment
5	High-value customer profile
6	Churn risk signals: behavioral combinations that predict cancellation

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Produced Features

Raw data was transformed into analysis-ready segments using Power Query. Key groupings include spend tiers, tenure lifecycle bands, support call buckets, activity levels, age groups, and days past due ranges. Two calculated columns drive the report's primary classifications: the Triple Crown flag (top spend + premium plan + long tenure) and the High-Risk flag (4+ support calls + 15+ days past due). Full logic and formulas are documented in the outro.

Why Power Query Instead of Excel

The dataset's size (~440,000 rows) exceeds Excel's reliable performance range for transformation and analysis. All cleaning and feature engineering was handled in Power Query using M language, then loaded into Power BI for modeling and visualization. The project was built on an Azure Virtual Machine to handle the memory demands of Power Query transformation and DAX evaluation at this scale.

Produced Features: Issue, Solution, Output

Issue: Raw age, tenure, support call counts, and spend values are continuous fields: useful for math, not for segmentation or visual grouping.

Solution: Conditional logic in Power Query to bucket each field into labeled, ordered groups.

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Feature	Input	Output
Age groups	Ranges: <25, 25–34, 35–44, 45–54, 55+	Young Adult → Senior
Tenure lifecycle	Months as customer, grouped	New → Established → Long-Term → Loyal
Support call bands	0 / 1–3 / 4–6 / 7–10	Ordinal risk tiers
Days past due buckets	0–5 / 6–10 / 11–15 / 15+	Delinquency tiers
Activity level	Derived from usage frequency field	Low / Medium / High
Spend tier	Percentile rank across all customers	Bottom 20% / Middle 60% / Top 20%

Issue: Power BI alphabetizes categorical axes by default, which breaks any segmentation that has a natural order (e.g., New → Loyal rendered as Loyal → Long-Term → New).

Solution: (S) sort-order helper columns paired with every (G) grouped field. Power BI's "Sort by Column" feature uses the (S) column to enforce correct ordering.

Naming convention:

- (G) = grouped/labeled field, used in visuals and slicers
- (S) = numeric sort-order column, hidden from report view

REVENUE ESTIMATION: METHODOLOGY & CONSTRAINTS

Why Create Revenue Estimates

The dataset provides cumulative lifetime spend -- not periodic revenue, not monthly billing, not margin. On its own, that number is hard to act on; for example, a customer with $2,000 in lifetime spend looks identical whether they've been a customer for 6 months or 48. To make spend data meaningful at a business level, it needed to be segmented, and projected.

Dataset Limits That Shaped the Approach

Tenure cap at 60 months Tenure is hard-capped at 60 in this dataset. There's no way to know whether lifetime spend continued accruing past that point, so 60-month accounts were excluded from all spend averages and revenue estimates. Including them would understate the true average monthly revenue rate for long-tenure customers.

Under-12-month accounts New accounts haven't had enough time to accrue meaningful lifetime spend. Their spend-to-tenure ratio is structurally lower than established customers -- not because they're lower value, but because they're newer. Including them in lifetime spend tier comparisons would pull averages down and misrepresent the segments. They're isolated into their own group within the spend tier structure.

What Was Built

Metric	Logic	Purpose
Estimated monthly revenue	Total spend ÷ tenure months (excludes 60-month and zero-spend accounts)	Normalize spend across tenures for fair comparison
Avg lifetime spend by segment	Average total spend within each tier (Top 20%, Middle 60%, Bottom 20%), 12–59 month accounts only	Understand true customer value by segment
Projected future revenue	Monthly rate × estimated remaining tenure	Forward-looking value per active customer
Revenue at risk	Projected revenue × churn probability by segment	Quantify what's lost if a segment churns

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Key DAX: Revenue Estimation

```
-- Average lifetime spend, Top 20% segment (tenure-normalized)
Top 20% Avg Customer Lifetime Total Spend =
CALCULATE(
    AVERAGE('Table'[Total Spend]),
    'Table'[(G) Lifetime Spend (12+ Months Tenure Accounts)] = "Top 20% Lifetime Spend",
    'Table'[Customer Tenure (Months)] < 60
)

-- Estimated monthly revenue per customer
Monthly Revenue (Estimated) =
IF(
```

```
  'Table'[Total Spend] > 0 &&
  'Table'[Customer Tenure (Months)] < 60,
  DIVIDE('Table'[Total Spend], 'Table'[Customer Tenure (Months)]),
  BLANK()
)
```

Calculated Columns: Key Flags

Triple Crown Customer Identifies the highest-value, most stable customer segment.

Criteria: Top 20% lifetime spend + Premium plan + Long-Term or Loyal tenure

```
Triple Crown =
IF(
 'Table'[Spend Tier] = "Top 20%" &&
 'Table'[Plan Type] = "Premium" &&
 'Table'[Tenure Group] IN {"Long-Term", "Loyal"},
 "Triple Crown",
 "Standard"
)
```

High-Risk Customer Identifies customers whose behavioral combination is strongly correlated with churn (~85% churn rate vs. ~22% baseline).

Criteria: 4 or more support calls AND 15 or more days past due

```
High Risk Flag =
IF(
 'Table'[Support Calls] >= 4 &&
 'Table'[Days Past Due] >= 15,
 "High Risk",
 "Standard"
)
```

Key DAX Measures

```
Churn Rate =
DIVIDE(
 COUNTROWS(FILTER('Table', 'Table'[Churn] = 1)),
 COUNTROWS('Table')
)

Revenue Contribution % =
DIVIDE(
 SUM('Table'[Total Spend]),
 CALCULATE(SUM('Table'[Total Spend]), ALL('Table'))
)

Support Burden % =
DIVIDE(
 SUM('Table'[Support Calls]),
```

```
  CALCULATE(SUM('Table'[Support Calls]), ALL('Table'))
)

Support Burden Index =
DIVIDE(
  [Support Burden %],
  [Revenue Contribution %]
)
```

Analytical Constraints

- No cost or margin data -- support call volume used as operational burden proxy
- Lifetime spend is cumulative, not periodic -- tenure-based normalization applied for comparability
- 30-day behavioral snapshot only. Findings are correlational, not causal or predictive
- Structural caps exist on certain fields (age ceiling, spend distribution, call count max, so segmentation at the extremes should be interpreted cautiously
- All derived metrics are directional estimates, not financial reporting figures

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Tools Used

Tool	Purpose
Power Query / M	Data cleaning, transformation, feature engineering
DAX	Measures, calculated columns, segmentation flags
Power BI Desktop	Modeling, report building, visualization
Azure Virtual Machine	Performance environment for large dataset processing
Kaggle	Dataset source